

Multilevel Models 2025/26

Model selection in multilevel (mixed-effects) models: metrics, criteria, and uncertainty

Danaia Burtseva

February 25, 2026

TU Dortmund University – Department of Statistics

What this presentation covers:

Three complementary papers:

- **Cantoni et al.:** how to properly quantify *goodness-of-fit* and *explained variation* in linear mixed models, showing that different metrics answer different questions and recommending a small set of reliable measures for practice.
- **Martínez-Huertas et al.:** when the true random-effects structure is unknown in crossed designs, how do different strategies affect inference on fixed effects?.
- **Müller et al.:** maps out the main families of model selection methods for linear mixed models and explains how they differ (IC, shrinkage, Fence, Bayesian).

Goal: summarize practical guidance for model selection in mixed models.

Setup: LMMs

Model setup: Linear mixed-effects model (LMM)

For groups $i = 1, \dots, m$:

$$\mathbf{y}_i = \mathbf{X}_i\boldsymbol{\beta} + \mathbf{Z}_i\mathbf{b}_i + \boldsymbol{\varepsilon}_i, \quad \mathbf{b}_i \sim \mathcal{N}(\mathbf{0}, \mathbf{D}), \quad \boldsymbol{\varepsilon}_i \sim \mathcal{N}(\mathbf{0}, \sigma^2\mathbf{I}_{n_i}).$$

$$\text{Var}(\mathbf{y}_i) = \mathbf{Z}_i\mathbf{D}\mathbf{Z}_i^\top + \sigma^2\mathbf{I}_{n_i}.$$

- Fixed effects $\boldsymbol{\beta}$: population-average relations
- Random effects $\mathbf{b}_i$: group-specific deviations (partial pooling)

Three key questions in mixed-model analysis

1) Model adequacy (fit / predictive quality)

- “Given a fitted LMM, how well does it describe the data?”
- Cantoni et al.: explained-variation and concordance measures

2) Model selection (choose among candidates)

- “Which model should we prefer, balancing fit and complexity?”
- Müller et al.: selection toolbox;

3) Model uncertainty

- “How sensitive is inference to the selected random structure?”
- Martínez-Huertas et al.: model selection vs model averaging and SE bias

Model adequacy (Cantoni et al.)

What exactly is model adequacy?

Model adequacy (Cantoni et al. 2024) evaluates how well a single fitted model describes or predicts the observed data.

Adequacy asks:

- Are predictions close to observations?
- Does the model capture key sources of variation?
- Is the remaining error “small” and structureless?

Why adequacy is tricky in LMMs (vs regression)

OLS regression:

$$R^2 = 1 - \frac{SS_{\text{res}}}{SS_{\text{tot}}}$$

A single variance decomposition is natural.

In an LMM:

$$y_i = X_i\beta + Z_i b_i + \varepsilon_i, \quad \text{Var}(y_i) = Z_i D Z_i^\top + \sigma^2 I_{n_i}.$$

Total variability has multiple sources:

- fixed effects (mean structure)
- random effects: $Z_i D Z_i^\top$ (between-group variation)
- residual noise: $\sigma^2 I_{n_i}$ (within-group variation)
- results depend on **marginal vs conditional** viewpoint

Marginal vs conditional adequacy (the key distinction)

Two prediction targets:

Marginal (fixed part only):

$$\hat{y}^{(m)} = X\hat{\beta}$$

- “How much do covariates explain at population level?”
- useful when discussing **fixed effects** (F-type adequacy)

Conditional (fixed + random):

$$\hat{y}^{(c)} = X\hat{\beta} + Z\hat{b}$$

- “How well do we predict for the *observed groups* after learning their random effects?”
- overall adequacy of the fitted mixed model (A-type adequacy)

Cantoni's classification: A vs F (adequacy)

Cantoni et al. classify adequacy measures by **what they credit**:

- **A: Overall adequacy** (fixed + random)
 - typically uses **conditional** predictions
- **F: Adequacy due to fixed effects** (fixed only)
 - typically uses **marginal** predictions

Practical meaning:

- A answers: "Is my mixed model fitting well overall?"
- F answers: "How much do the *fixed covariates* contribute?"

Relative vs absolute adequacy (null model issue)

Some adequacy measures are **relative**:

- they compare your model to a **null/baseline** model
- e.g., proportional reduction in deviance/variance vs intercept-only

Implication:

- changing the null model can change the value and interpretation
- always state the baseline if you report a relative measure

Concordance measures evaluate agreement between observations and predictions.

High concordance means:

- predictions track the data well
- both mean and scale are reasonable

Why useful in LMMs:

- works for both marginal and conditional predictions
- does not require a null model

Adjusted concordance coefficient

Cantoni et al. highlight adjusted concordance (e.g., $r_{c,a}$) as an attractive adequacy summary. The adjusted concordance coefficient combines covariance and variance terms to penalize both bias and scale mismatch.

$$r_{c,a} = \frac{2 \widehat{\text{Cov}}(y, \hat{y})}{\widehat{\text{Var}}(y) + \widehat{\text{Var}}(\hat{y}) + (\bar{y} - \bar{\hat{y}})^2}.$$

Interpretation:

- $r_{c,a} \approx 1 \rightarrow$ strong agreement
- $r_{c,a} \approx 0 \rightarrow$ weak agreement
- can be negative if predictions are systematically biased

Penalized explained variation: why penalize?

Standard R^2 -type measures tend to increase as model flexibility grows.

In LMMs, adding random effects often reduces deviance:

$$D_{\text{model}} \downarrow \Rightarrow R^2_{\text{like}} \uparrow$$

- improvement may reflect flexibility rather than true structure
- naive explained variation can therefore be over-optimistic

Penalized explained variation (Cantoni et al.)

Cantoni et al. consider a penalized proportional reduction in deviance:

$$P_{\text{rand}}^{\text{pen}} = 1 - \frac{D_{\text{model}} + \kappa}{D_{\text{null}}},$$

where

- D_{model} = deviance of the fitted mixed model
- D_{null} = deviance of the baseline model
- κ = penalty for model flexibility

Interpretation:

- reduces artificial improvement from complex random effects
- provides a more conservative adequacy summary

Pooling / shrinkage in mixed models

In an LMM with random intercepts:

$$y_{ij} = \mathbf{x}_{ij}^\top \beta + b_i + \varepsilon_{ij}, \quad b_i \sim \mathcal{N}(0, \sigma_b^2), \quad \varepsilon_{ij} \sim \mathcal{N}(0, \sigma^2).$$

The BLUP for the group effect has the form

$$\hat{b}_i \approx \underbrace{\frac{\sigma_b^2}{\sigma_b^2 + \sigma^2/n_i}}_{\text{shrinkage factor}} (\bar{y}_i - \mathbf{x}_i^\top \hat{\beta}).$$

- small n_i or large noise $\sigma^2 \Rightarrow$ stronger shrinkage
- large between-group variance $\sigma_b^2 \Rightarrow$ weaker shrinkage

Why this matters for adequacy:

- strong pooling stabilizes predictions (often better adequacy)
- weak pooling can overfit small groups

Pooling factors (Gelman & Pardoe) summarize the strength of hierarchical pooling.

What Cantoni recommend to report

Cantoni's message is **not** to report many competing measures.

A defensible minimal set:

- **Overall adequacy (A):** conditional P_{rand} and/or conditional $r_{c,a}$
- **Fixed-effect adequacy (F):** marginal versions of these measures

Interpretation:

- $P_{\text{rand}} \rightarrow$ proportional improvement (“ R^2 -like”)
- $r_{c,a} \rightarrow$ predictive agreement

Model selection (Müller et al.)

Why model selection is harder in LMMs

Compared with ordinary regression, mixed models introduce additional challenges:

- **Dependent observations:** data within the same cluster are correlated
- **Two model layers:** both fixed effects and random-effects structure may need selection
- **Boundary issues:** testing random effects involves variance parameters at $\sigma_b^2 = 0$
- **Hidden flexibility:** random effects increase model complexity through shrinkage

Key message (Müller et al. 2013): model complexity in LMMs is not just the number of parameters.

Practical guidance (Müller et al.)

Choice of selection method should match the goal.

Goal	Reasonable tools
Fixed-effect selection	mAIC/BIC, penalization methods
Random-structure selection	cAIC, careful IC comparison
Conditional prediction	cAIC-style criteria
Many candidate effects	shrinkage or Bayesian methods

No single method dominates in all LMM settings.

Random-effects structure

In mixed models, random effects behave like **partially pooled (data-adaptive) parameters**.

For conditional prediction,

$$\hat{y} = X\hat{\beta} + Z\hat{b},$$

the flexibility of the fitted model is measured by the **effective degrees of freedom**:

$$\rho(\tau) = \text{trace}\{H(\tau)\}.$$

- $p \leq \rho(\tau) \leq p + s$
- increases as random effects are less strongly shrunk

Implication for cAIC:

$$\text{penalty} = 2\rho(\tau) \quad \text{instead of} \quad 2k.$$

Model uncertainty (Martínez-Huertas)

Crossed random effects in experiments

Many experiments contain two **crossed sources of variability**:

- **Subjects:** individuals differ in their overall response level
- **Items:** stimuli differ in difficulty or intensity

A common crossed mixed model is:

$$y_{si} = X_{si}\beta + u_{0s} + u_{0i} + \varepsilon_{si},$$

- $X_{si}\beta$: population mean structure (fixed effects)
- u_{0s} : subject-specific random intercept
- u_{0i} : item-specific random intercept

Key issue: the chosen random-effects structure can strongly affect standard errors and inference.

What the paper compare

Martínez-Huertas et al. (Martínez-Huertas et al. 2021) study two strategies for random-structure uncertainty.

Model selection:

- likelihood-ratio tests (LRT)
- information criteria (AIC, BIC)
- bottom-up vs top-down search

Model averaging:

- compute Akaike weights across candidate models
- average standard errors of fixed effects

Goal: assess consequences for inference under model uncertainty.

Main outcome: bias of fixed-effect SEs

Primary evaluation metric (Martínez-Huertas et al.):

$$\text{SE bias} = 100 \cdot \frac{\widehat{SE} - SD(\hat{\theta})}{SD(\hat{\theta})}.$$

- $\widehat{SE}$: model-based standard error
- $SD(\hat{\theta})$: empirical SD across simulations (treated as truth)

Interpretation:

- negative bias $\Rightarrow$ SE underestimated
- positive bias $\Rightarrow$ SE overestimated

Main outcome: bias of fixed-effect SEs

Main drivers of performance:

- number of subjects
- number of items
- magnitude of random slopes

Key message:

- under-parameterizing the random structure often underestimates SEs
- model averaging and model selection show broadly similar SE behavior

Practical workflow for mixed-model analysis

Tip 1: Select the model carefully (Müller et al.)

- match the criterion to the prediction target
- use cAIC for conditional prediction / random-structure decisions
- avoid relying on a single automatic rule

Tip 2: Check model adequacy (Cantoni et al.)

- report both marginal and conditional adequacy
- use a small, interpretable set (e.g., P_{rand} , $r_{c,a}$)

Tip 3: Assess model uncertainty (Martínez-Huertas et al.)

- check sensitivity to the random-effects specification
- consider model averaging when the structure is uncertain

References

-  Cantoni, Eva, Nicolas Jacot, and Paolo Ghisletta (2024). **“Review and comparison of measures of explained variation and model selection in linear mixed-effects models”**. In: *Econometrics and Statistics* 29, pp. 150–168. DOI: 10.1016/j.ecosta.2023.03.002.
-  Martínez-Huertas, José A., Raúl Olmos, and Emilio Ferrer (2021). **“Model Selection and Model Averaging for Mixed-Effects Models with Crossed Random Effects for Subjects and Items”**. In: *Multivariate Behavioral Research* 56.6, pp. 884–905. DOI: 10.1080/00273171.2020.1763404.
-  Müller, Samuel, J. L. Scealy, and A. H. Welsh (2013). **“Model Selection in Linear Mixed Models”**. In: *Statistical Science* 28.2, pp. 135–167. DOI: 10.1214/12-STS410.